Automatic Segmentation and Structural Analysis of Humpback Whale Song

Student Information

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1 Project Overview

Humpback whale song is one of the most complex acoustic displays in the animal kingdom, organised hierarchically into units, phrases, and themes that combine into a repeating song cycle. Marine biologists currently rely heavily on manual, time-consuming annotation to study this structure, which limits the scale at which song can be analysed across populations and years. This project aims to build a signal-processing pipeline that automatically segments humpback whale song recordings into their constituent units and phrases, and identifies recurring phrase patterns without relying on hand-labelled training data. The proposed solution combines short-time spectral analysis with unsupervised sequence pattern discovery to recover the hierarchical structure of a song directly from audio, producing an output that can be compared against existing bioacoustic annotations.

2 Background and Motivation

Humpback whale song was first formally described by Payne and McVay (1971), who observed that songs are built from repeating sequences of sound units and evolve gradually over time within a population. Since then, most large-scale studies of song structure have depended on manual spectrogram annotation by trained analysts, a process that does not scale well to the large hydrophone archives now available from long-term ocean monitoring networks. Existing automated approaches typically use supervised classifiers trained on labelled unit types, which perform well within a single dataset but generalise poorly across different populations, recording conditions, and years, since whale song itself changes over time. This project is motivated by the appeal of an unsupervised, more general approach: rather than training a classifier to recognise fixed unit categories, the pipeline will discover repeating structural patterns directly from the acoustic data. This topic was chosen because it sits at the intersection of classical signal processing and sequence analysis, and because publicly available annotated datasets make it possible to properly evaluate the results.

3 Proposed Methodology

The project will be implemented in MATLAB, using the Signal Processing Toolbox and Audio Toolbox, with the following pipeline:

- **Pre-processing:** Band-pass filtering (`bandpass` / `designfilt`) and normalisation of raw hydrophone recordings to isolate the frequency range of humpback vocalisations (roughly 20 Hz–10 kHz).

- **Time-frequency representation:** Short-time Fourier transform (`stft` / `spectrogram`) to generate spectrograms, with parameters tuned to the duration of individual song units.
- **Unit segmentation:** Energy-based onset/offset detection on the spectrogram (adaptive-threshold envelope detection) to isolate individual sound units under varying background noise.
- **Feature extraction:** Representation of each segmented unit using spectral features such as mel-frequency cepstral coefficients (`mfcc`), spectral centroid (`spectralCentroid`), and duration.
- **Pattern discovery:** Clustering of unit features (`kmeans` or `clusterdata` for hierarchical clustering) to assign discrete symbols to units, followed by sequence analysis (custom n-gram/motif-discovery routines) to identify recurring phrases and candidate themes.
- **Evaluation:** Comparison of automatically discovered structure against existing annotated song transcriptions.

Data sources will primarily be drawn from the Watkins Marine Mammal Sound Database and, if access permits, humpback song archives referenced by MobySound, both of which provide recordings with varying degrees of existing annotation suitable for evaluation.

4 Expected Outcomes

- A working MATLAB pipeline that takes a raw humpback whale recording and outputs a segmented, labelled sequence of units and candidate phrases.
- Quantitative evaluation of segmentation accuracy (e.g. unit boundary precision/recall) against annotated reference recordings.
- A qualitative and quantitative comparison of discovered recurring phrase patterns against known song structure from prior bioacoustic literature.
- Full documentation and reproducible MATLAB code (scripts/live scripts) hosted on GitHub, including example spectrograms and segmentation outputs.

5 Timeline (Weeks 1–13)

Week	Task
1–2	Finalise project scope; identify and download suitable datasets
3–5	Literature review of whale bioacoustics and unsupervised sequence discovery methods
6–9	Implement pre-processing, STFT analysis, and unit segmentation pipeline
10–11	Implement feature extraction and phrase-pattern clustering; evaluate against annotations
12–13	Refine results, write final report, and complete GitHub documentation

6 References

1. Payne, R. S., & McVay, S. (1971). Songs of humpback whales. *Science*, 173(3997), 585–597.
2. Mercado III, E. (2018). Acoustic signaling by humpback whales (*Megaptera novaeangliae*). *Aquatic Mammals*, 44(2), 186–204.

3. Suzuki, R., Buck, J. R., & Tyack, P. L. (2006). Information entropy of humpback whale songs. *The Journal of the Acoustical Society of America*, 119(3), 1849–1866.

4. Malige, F., Patris, J., & Adam, O. (2021). A deep learning approach for the analysis of humpback whale song units. *Applied Sciences*, 11(21), 9756.

7 Appendix

Preliminary spectrogram examples and initial exploratory segmentation results will be added here once dataset access is confirmed and early implementation is underway.